nature are transmitted at the quantum level by other particles, the messenger particles, which are summarized in Table 12.2**.** String theory accounts for the messenger particles exactly as it accounts for the matter particles. Namely, each messenger particle is a string that's executing a particular vibrational pattern. A photon is a string vibrating in one particular pattern, a W particle is a string vibrating in a different pattern, a gluon is a string vibrating in yet another pattern. And, of prime importance, what Schwarz and Scherk showed in 1974 is that there is a particular vibrational pattern that has all the properties of a graviton, so that the gravitational force is included in string theory's quantum mechanical framework. Thus, not only do matter particles arise from vibrating strings, but so do the messenger particles—even the messenger particle for gravity.

Force	Force particle	Mass
Strong	Gluon	0
Electromagnetic	Photon	0
Weak	W, Z	86, 97
Gravity	Graviton	0

Table 12.2 The four forces of nature, together with their associated force particles and their masses in multiples of the proton mass. (There are actually two W particles—one with charge +1 and one with charge −1—that have the same mass; for simplicity we ignore this detail and refer to each as aW particle.

And so, beyond providing the first successful approach for merging gravity and quantum mechanics, string theory revealed its capacity to provide a unified description of all matter and all forces. That's the claim that knocked thousands of physicists off their chairs in the mid-1980s; by the time they got up and dusted themselves off, many were converts.

Why Does String Theory Work?

Before the development of string theory, the path of scientific progress was strewn with unsuccessful attempts to merge gravity and quantum